Stimulus-Construction Protocol

Recognition-Oriented vs. Empathic Communication

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Supplementary Materials

Purpose

This protocol defines the operationalization of communicative cues used to construct chatbot messages across experimental conditions. Its primary goal is to ensure internal validity, conceptual clarity, and a clear separation between recognition-oriented and empathic communication in a scripted NGO AI chatbot interaction. The protocol was finalized prior to stimulus writing and applied consistently across all conditions. All stimuli are based on a constant informational core[1], combined with condition-specific language layers. For the recognition-oriented layer, theoretical ideas of recognition were translated into observable institutional communication features that can be experimentally manipulated. Differences between conditions must be restricted to the communicative cues defined below.

1. Conceptual Framework

This experiment distinguishes between two independent communicative dimensions and enforces a strict separation based on the following logic:

1.1 Recognition-Oriented Communication (High-R)

Drawing on recognition theory (Honneth, 1995; Fraser, 2003), recognition-oriented communication is conceptualized as an institutional communication logic that structures the user–institution relationship through explicit boundary cues. It positions the user as a competent and decision-capable participant within a structured interaction. Implemented through the cues illustrated below, its defining feature lies in how these cues jointly signal user decision authority, legitimate refusal, and bounded institutional roles. This allows it to be examined as a distinct and empirically testable framing dimension, separate from empathic communication. Recognition operates through non-affective communication and functions as a boundary-setting mechanism that defines the scope of the interaction without increasing relational closeness.

1.2 Empathic Communication (High-E)

Empathic communication is operationalized as the use of affective and interpersonal language cues that signal emotional understanding, validation, and relational warmth. These cues reflect the communicative performance of empathy commonly observed in contemporary conversational AI systems and may include mild socially oriented language that increases perceived interpersonal engagement (e.g. De Visser et al., 2020). Empathic communication operates exclusively in the affective-relational domain. For experimental control, it is deliberately decoupled from institutional boundary-setting and does not communicate role clarity, procedural purpose, or user decision authority.

[1] See supplementary material «Stimuli and Scenario", Condition 1

2. Operationalization: Markers and Constraints

2.1 Dimension 1: Recognition-Oriented Communication (R)

Goal: Signal procedural legitimacy, role clarity, user autonomy, and equal standing.

Component	***Definition***	***Markers (Examples)***	***Anti-Markers (Exclusions)***
R1: Decision Authority	Explicitly locating the decision with the user.	• "You decide how to proceed." • "It is up to you whether you want to take this step." • "You can choose which option fits best for you"	• *Imperatives* ("You must", "please do", "you should proceed like this") • *Pseudo-options* ("you can do X, but Y would actually be better") • *Authority claims / Steering* ("I strongly recommend", "the best option is")
R2: Legitimate Refusal	Framing non-participation as a valid option.	• "Choosing not to share this is a valid option." • "You may also continue without providing it" • "Saying no is a normal option here"	• *Guilt-framing* ("in order to help you, we need...") • *Moral pressure* ("if you really want support…") • *Implicit sanctions* ("without X, I cannot help") without offering an alternative
R3: Procedural Transparency	Clarifying the functional 'why' of a request.	• "I am asking this to identify suitable options for you." • "This conversation serves as an initial orientation." • "You can also decide to clarify this later."	• *Unexplained requests* ("Please provide X.") • *Functional opacity* ("I need this now.") • Implicit data use without stated purpose
R4: Role Clarity	Defining the chatbot's mandate and limitations.	• "I can provide information and outline options, but I do not decide for you." • "I cannot make any binding decisions." • "I do not replace personal consultation but can support initial orientation."	• *Assumption of responsibility* ("I will take care of this.") • *Solution guarantees* ("We will get this sorted.") • *Substitution of human actors* ("I will do this for you")
R5: Status Parity	Addressing the user as a competent agent.	• Neutral, professional tone • No assumptions of deficit • No simplifications that imply limited competence • No emotional attributions without explicit user cues	• *Pity expressions* without trigger ("I understand how hard this must be for you") • *Infantilizing language* ("don't worry," "this is very simple.") • *Assumptions of incapacity* ("people in your situation often struggle to…")

Sources: *The operationalization draws on Fraser's (2003) and Honneth's (1995) models of recognition, conceptualizing recognition as the attribution of equal standing and legitimate participation. This framework is complemented by insights from Self-Determination Theory (Ryan & Deci, 2017) to specify agency attribution, as well as the dark pattern taxonomy of Gray et al. (2018) to define exclusion criteria and institutional framing elements (see Eslami et al., 2018; Suchman, 2007).*

Constraint (R): Recognition cues must remain functional in tone, concise, and non-affective. They must not imply any phrasing that could be read as emotional support, reassurance, or an attempt to create relational closeness.

2.2 Dimension 2: Empathic Communication (E)

Goal: Signal emotional understanding and interpersonal responsiveness.

Component	***Definition***	***Markers (Examples)***	***Anti-Markers (Exclusions)***
E1: Emotional Acknowledgment	Recognizing potential stress or uncertainty.	• "This can feel like a stressful situation." • "Many people experience this situation as stressful."	• *Attribution of decision authority* (e.g., "you decide…") • *Role clarification* or *mandate statement* ("My function is...") • *Procedural explanation* ("This conversation serves as…")
E2: Validation	Confirming that the user's feelings are reasonable.	• "It is understandable that you feel unsure." • "It makes sense in a situation like this."	• *Decision guidance* ("You should proceed like this") • *Institutional framing*
E3: Interpersonal Warmth	Mild socially responsive language that signals attentiveness and interpersonal warmth.	• Soft conversational tone • "Thank you for sharing that." • Use of subtle emojis.	• *Strong reassurance* ("you are not alone") • *Care commitments* ("I am here all the time") • *Therapeutic language*
E4: Perspective-Taking	Framing the situation from the user's viewpoint.	• "From your perspective, this might raise questions."	• *Strong empathy claims* ("I know how exactly how you feel.") • *Strong co-action*

Note: *Examples reflect common patterns observed in contemporary conversational AI systems.*

Constraint (E): Empathy cues must remain affective and non-institutional. They must not communicate authority, boundaries, or procedural control.

3. Stimulus Engineering & Orthogonality Rules

To ensure experimental control, chatbot responses are constructed using a layered design approach (see De Gennaro et al., 2020; Lahnala, 2025). Each message is composed of a constant functional core and variable layers.

3.1 The Layered Construction Logic

1. Constant Core: Contains the factual information or task-related response. It is strictly neutral and devoid of any R- or E-cues.
2. High-R Layer: Adds markers from R1-R5.
3. High-E Layer: Adds markers from E1-E4.
4. Low-R / Low-E: The absence of the respective markers (unmarked interaction).

In the High-E/High-R condition, both layers are combined additively without introducing any additional stylistic or functional elements beyond the specified markers.

3.2 Practical Application Rules

- Length Control: High-R implementation must not lead to significantly longer messages. Recognition is implemented via short, functional sentences.
- Tone Control: Differences in politeness, conversational smoothness, or general friendliness beyond the specified R- and E-markers are minimized across conditions.
- Exclusionary Mapping: In a High-R/Low-E condition, any phrasing that could be interpreted as emotional warmth is strictly prohibited. Conversely, in High-E/Low-R, any phrasing that clarifies user authority or system boundaries is removed.
- Any cue that simultaneously communicates affective warmth and institutional authority is excluded.

4. References

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